



Image-to-image Translation models introduction

Lab Team

I2I Translation models

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Supervised Learning
(requires paired training data)

Uni-modal
(1 input - 1 output)

- Pix2Pix (2016): [paper](#)
- Pix2PixHD (2017): [paper](#)
- SPADE (2019): [paper](#)

Multi-modal
(1 input - N output)

- BicycleGAN (2017): [paper](#)

Unsupervised Learning
(requires unpaired training data)

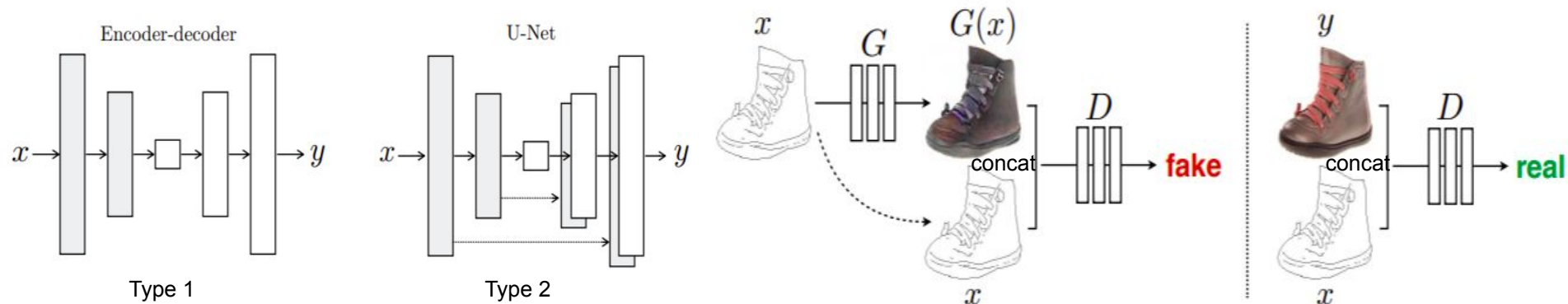
Uni-modal
(1 input - 1 output)

- CycleGAN (2017): [paper](#)
- UNIT (2017): [paper](#)
- StarGAN (2017): [paper](#)
- DGNet (2019): [paper](#)

Multi-modal
(1 input - N output)

- MUNIT (2018): [paper](#)
- DRIT (2018): [paper](#)
- DRIT++ (2019): [paper](#)
- StarGANv2 (2019): [paper](#)
- FUNIT(2019): [paper](#)

Pix2Pix (Nov 2016)



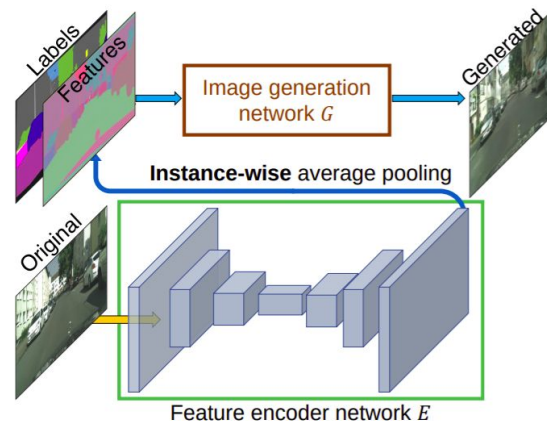
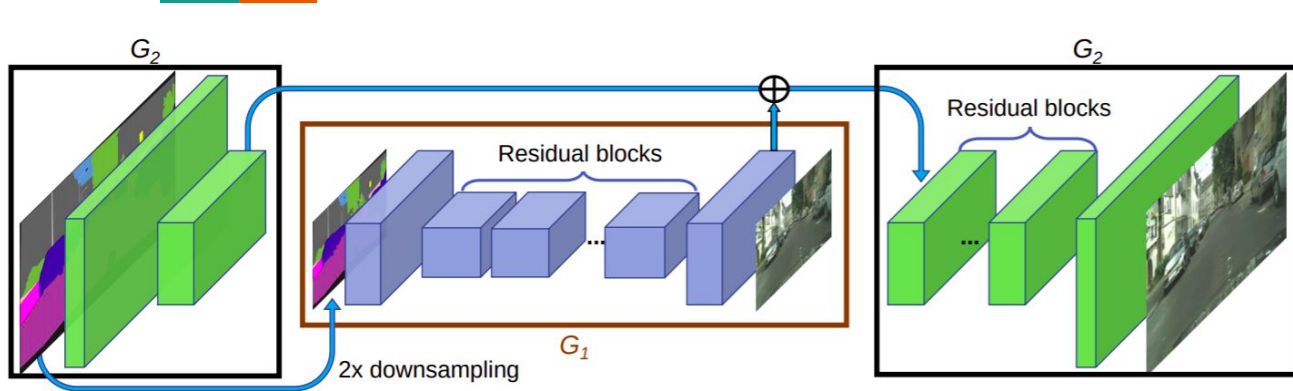
$$\mathcal{L}_{cGAN}(G, D) = \mathbb{E}_{x,y}[\log D(x, y)] + \mathbb{E}_{x,z}[\log(1 - D(x, G(x, z)))]$$

$$\mathcal{L}_{L1}(G) = \mathbb{E}_{x,y,z}[\|y - G(x, z)\|_1].$$

(Author used L1 instead of L2 because L1 encourages less blurring)

(Although Pix2Pix had input noise z but the model seemed to ignore it and produced deterministic output. In the final experiment, author removed noise z and used Dropout instead but they observed only minor stochasticity in the output)

Pix2PixHD (Nov 2017)

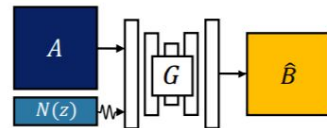
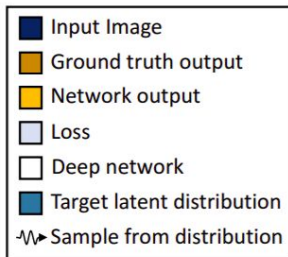


$$\mathcal{L}_{\text{FM}}(G, D_k) = \mathbb{E}_{(\mathbf{s}, \mathbf{x})} \sum_{i=1}^T \frac{1}{N_i} [\|D_k^{(i)}(\mathbf{s}, \mathbf{x}) - D_k^{(i)}(\mathbf{s}, G(\mathbf{s}))\|_1],$$

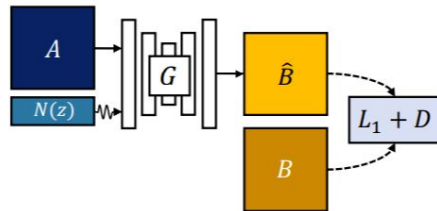
$$\min_G \left(\left(\max_{D_1, D_2, D_3} \sum_{k=1,2,3} \mathcal{L}_{\text{GAN}}(G, D_k) \right) + \lambda \sum_{k=1,2,3} \mathcal{L}_{\text{FM}}(G, D_k) \right)$$

All discriminators were used as feature extractors in Feature Matching Loss ($\mathcal{L}_{\text{FM}}$) and weren't optimized to maximize this loss

BicycleGAN (Nov 2017)



(a) Testing Usage for all models



(b) Training pix2pix+noise

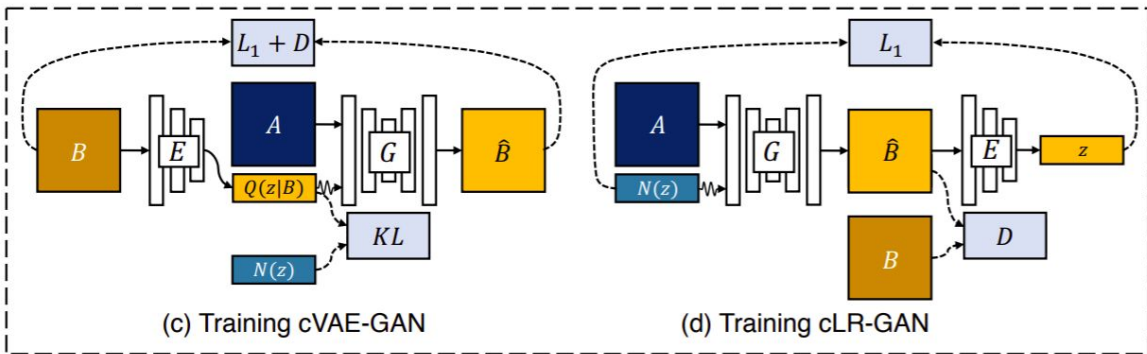
$$\mathcal{L}_{\text{GAN}}(G, D) = \mathbb{E}_{\mathbf{A}, \mathbf{B} \sim p(\mathbf{A}, \mathbf{B})} [\log(D(\mathbf{A}, \mathbf{B}))] + \mathbb{E}_{\mathbf{A} \sim p(\mathbf{A}), \mathbf{z} \sim p(\mathbf{z})} [\log(1 - D(\mathbf{A}, G(\mathbf{A}, \mathbf{z})))]$$

$$\mathcal{L}_1^{\text{latent}}(G, E) = \mathbb{E}_{\mathbf{A} \sim p(\mathbf{A}), \mathbf{z} \sim p(\mathbf{z})} \|\mathbf{z} - E(G(\mathbf{A}, \mathbf{z}))\|_1$$

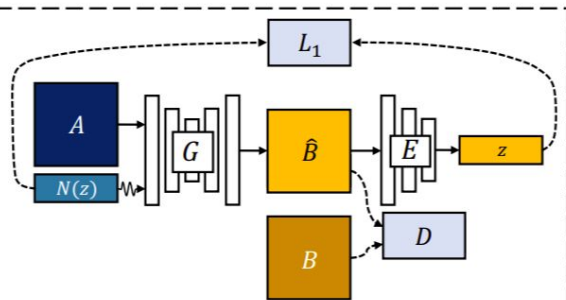
$$\mathcal{L}_{\text{KL}}(E) = \mathbb{E}_{\mathbf{B} \sim p(\mathbf{B})} [\mathcal{D}_{\text{KL}}(E(\mathbf{B}) \| \mathcal{N}(0, I))],$$

$$\mathcal{L}_1^{\text{VAE}}(G) = \mathbb{E}_{\mathbf{A}, \mathbf{B} \sim p(\mathbf{A}, \mathbf{B}), \mathbf{z} \sim E(\mathbf{B})} \|\mathbf{B} - G(\mathbf{A}, \mathbf{z})\|_1$$

$$\mathcal{L}_{\text{GAN}}^{\text{VAE}} = \mathbb{E}_{\mathbf{A}, \mathbf{B} \sim p(\mathbf{A}, \mathbf{B})} [\log(D(\mathbf{A}, \mathbf{B}))] + \mathbb{E}_{\mathbf{A}, \mathbf{B} \sim p(\mathbf{A}, \mathbf{B}), \mathbf{z} \sim E(\mathbf{B})} [\log(1 - D(\mathbf{A}, G(\mathbf{A}, \mathbf{z})))]$$



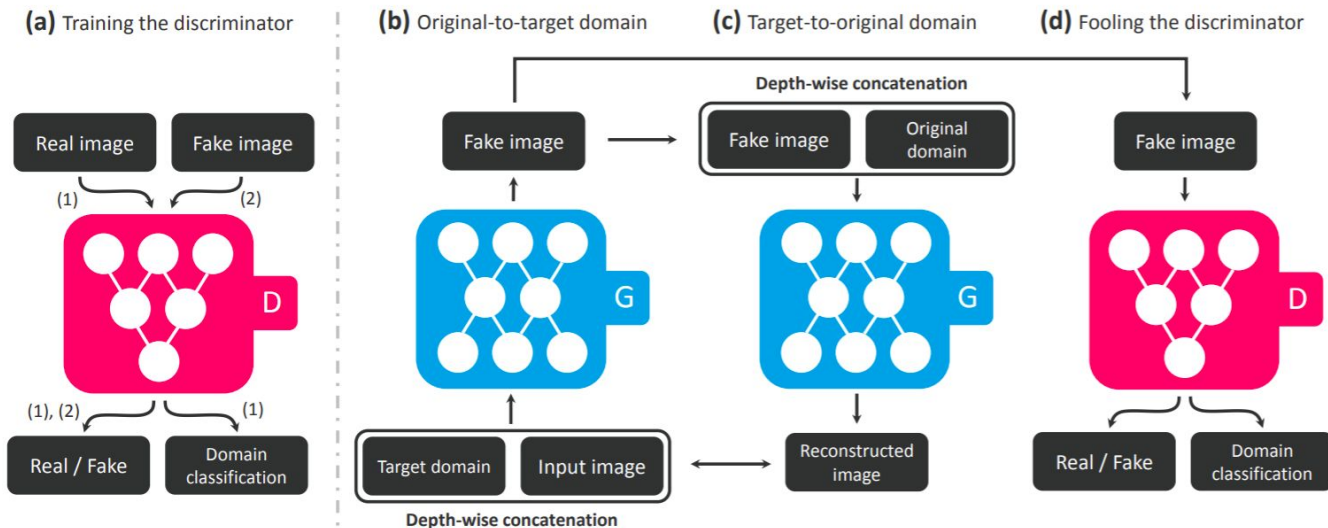
(c) Training cVAE-GAN



(d) Training cLR-GAN

(e) Training BicycleGAN

StarGAN (Nov 2017)



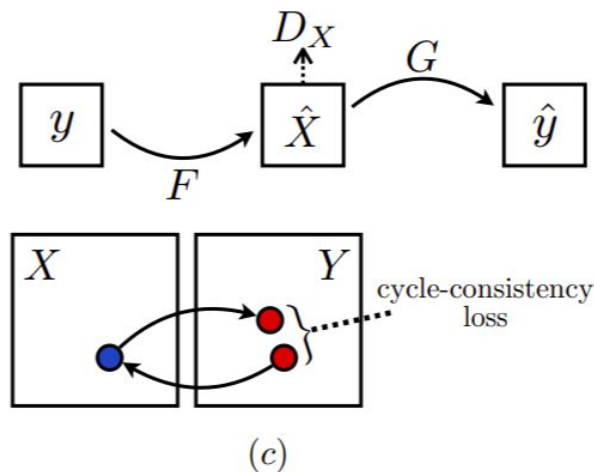
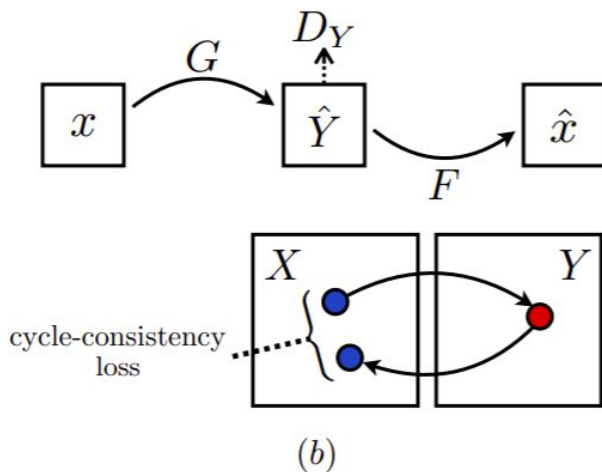
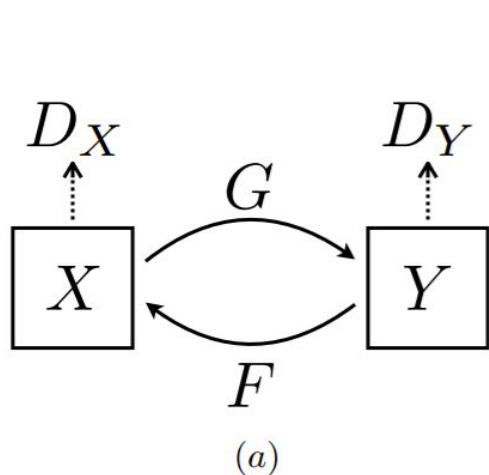
$$\mathcal{L}_{adv} = \mathbb{E}_x [\log D_{src}(x)] + \mathbb{E}_{x,c} [\log (1 - D_{src}(G(x, c)))], \quad (1)$$

$$\mathcal{L}_{cls}^r = \mathbb{E}_{x,c'} [-\log D_{cls}(c'|x)], \quad (2)$$

$$\mathcal{L}_{cls}^f = \mathbb{E}_{x,c} [-\log D_{cls}(c|G(x, c))]. \quad (3)$$

$$\mathcal{L}_{rec} = \mathbb{E}_{x,c,c'} [\|x - G(G(x, c), c')\|_1], \quad (4)$$

CycleGAN (Mar 2017)



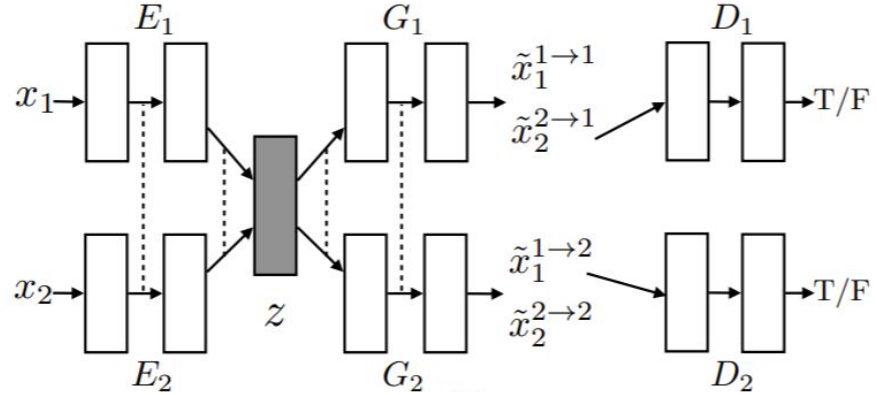
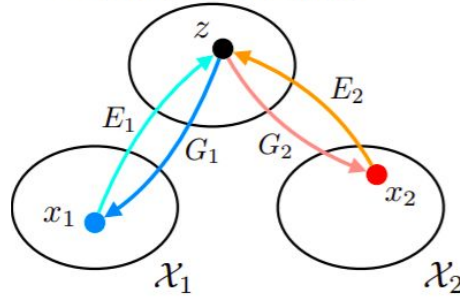
$$\mathcal{L}_{\text{GAN}}(G, D_Y, X, Y) = \mathbb{E}_{y \sim p_{\text{data}}(y)} [\log D_Y(y)] \\ + \mathbb{E}_{x \sim p_{\text{data}}(x)} [\log(1 - D_Y(G(x)))] ,$$

$$\mathcal{L}_{\text{GAN}}(F, D_X, Y, X) = \dots$$

$$\mathcal{L}_{\text{cyc}}(G, F) = \mathbb{E}_{x \sim p_{\text{data}}(x)} [\|F(G(x)) - x\|_1] \\ + \mathbb{E}_{y \sim p_{\text{data}}(y)} [\|G(F(y)) - y\|_1] .$$

UNIT (Mar 2017)

$\mathcal{Z}$: shared latent space



$$\mathcal{L}_{\text{GAN}_1}(E_2, G_1, D_1) = \lambda_0 \mathbb{E}_{x_1 \sim P_{\mathcal{X}_1}} [\log D_1(x_1)] + \lambda_0 \mathbb{E}_{z_2 \sim q_2(z_2|x_2)} [\log(1 - D_1(G_1(z_2)))]$$

$$\mathcal{L}_{\text{GAN}_2}(E_1, G_2, D_2) = \lambda_0 \mathbb{E}_{x_2 \sim P_{\mathcal{X}_2}} [\log D_2(x_2)] + \lambda_0 \mathbb{E}_{z_1 \sim q_1(z_1|x_1)} [\log(1 - D_2(G_2(z_1)))]$$

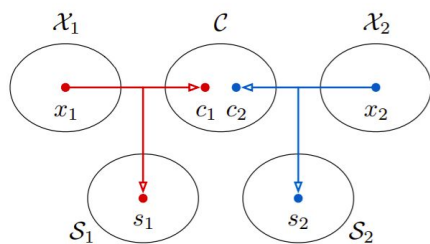
$$\mathcal{L}_{\text{VAE}_1}(E_1, G_1) = \lambda_1 \text{KL}(q_1(z_1|x_1) || p_\eta(z)) - \lambda_2 \mathbb{E}_{z_1 \sim q_1(z_1|x_1)} [\log p_{G_1}(x_1|z_1)]$$

$$\mathcal{L}_{\text{VAE}_2}(E_2, G_2) = \lambda_1 \text{KL}(q_2(z_2|x_2) || p_\eta(z)) - \lambda_2 \mathbb{E}_{z_2 \sim q_2(z_2|x_2)} [\log p_{G_2}(x_2|z_2)]$$

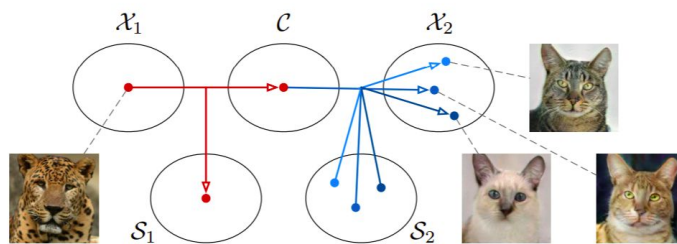
$$\mathcal{L}_{\text{CC}_1}(E_1, G_1, E_2, G_2) = \lambda_3 \text{KL}(q_1(z_1|x_1) || p_\eta(z)) + \lambda_3 \text{KL}(q_2(z_2|x_1^{1 \rightarrow 2}) || p_\eta(z)) - \lambda_4 \mathbb{E}_{z_2 \sim q_2(z_2|x_1^{1 \rightarrow 2})} [\log p_{G_1}(x_1|z_2)]$$

$$\mathcal{L}_{\text{CC}_2}(E_2, G_2, E_1, G_1) = \lambda_3 \text{KL}(q_2(z_2|x_2) || p_\eta(z)) + \lambda_3 \text{KL}(q_1(z_1|x_2^{2 \rightarrow 1}) || p_\eta(z)) - \lambda_4 \mathbb{E}_{z_1 \sim q_1(z_1|x_2^{2 \rightarrow 1})} [\log p_{G_2}(x_2|z_1)].$$

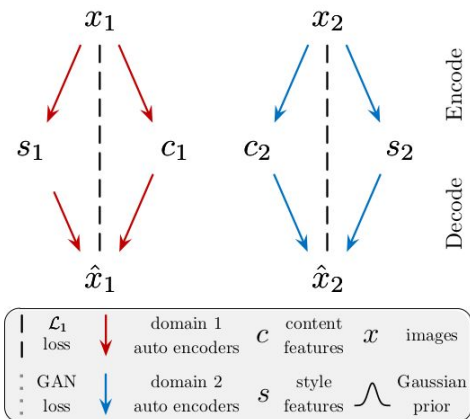
MUNIT (Apr 2018)



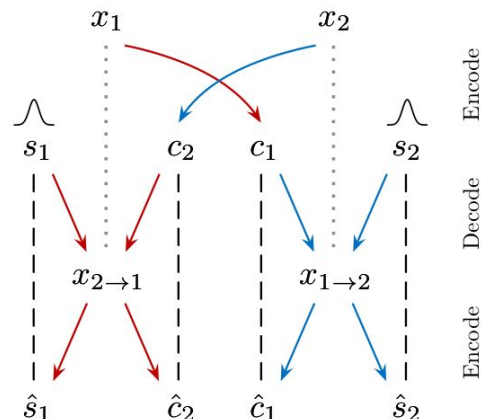
(a) Auto-encoding



(b) Translation

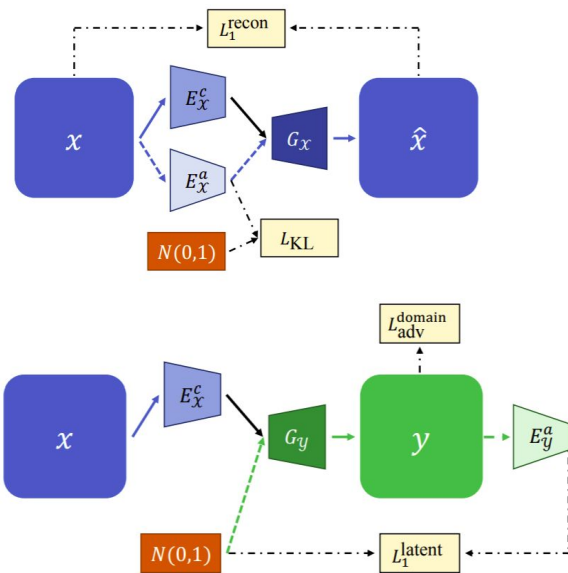
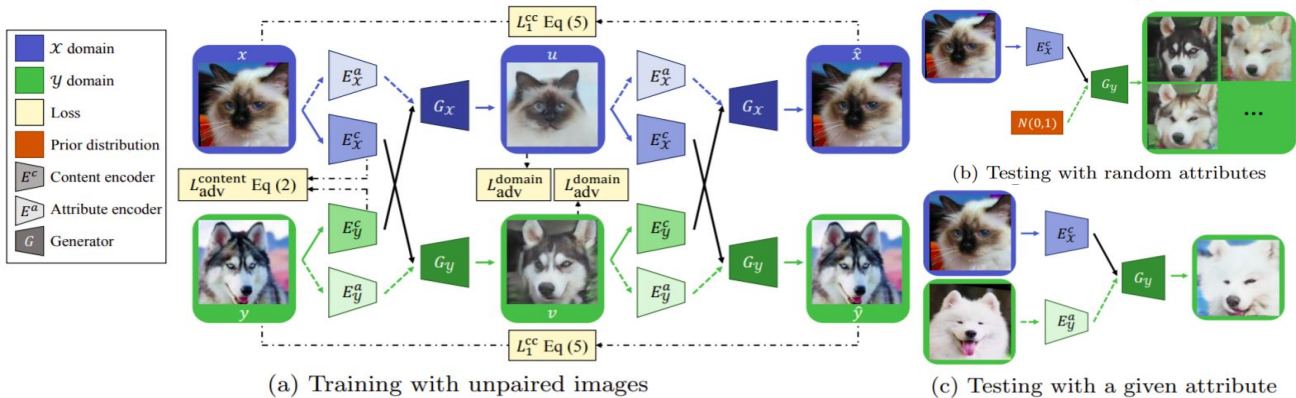


(a) Within-domain reconstruction



(b) Cross-domain translation

DRIT (Aug 2018)

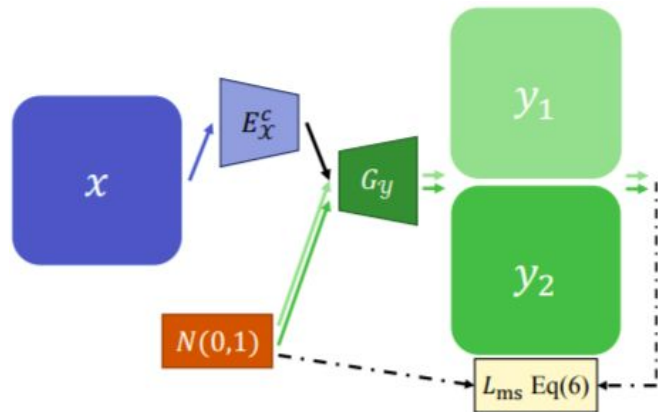


$$L_{adv}^{content}(E_X^c, E_Y^c, D^c) = \mathbb{E}_x \left[\frac{1}{2} \log D^c(E_X^c(x)) + \frac{1}{2} \log (1 - D^c(E_X^c(x))) \right] \\ + \mathbb{E}_y \left[\frac{1}{2} \log D^c(E_Y^c(y)) + \frac{1}{2} \log (1 - D^c(E_Y^c(y))) \right]$$

$$L_1^{cc}(G_X, G_Y, E_X^c, E_Y^c, E_X^a, E_Y^a) = \mathbb{E}_{x,y} [\|G_X(E_Y^c(v), E_X^a(u)) - x\|_1 \\ + \|G_Y(E_X^c(u), E_Y^a(v)) - y\|_1],$$

$$\min_{G, E^c, E^a} \max_{D, D^c} \lambda_{adv}^{content} L_{adv}^c + \lambda_1^{cc} L_1^{cc} + \lambda_{adv}^{domain} L_{adv}^{domain} + \lambda_1^{recon} L_1^{recon} \\ + \lambda_1^{latent} L_1^{latent} + \lambda_{KL} L_{KL}$$

DRIT++ (May 2019)

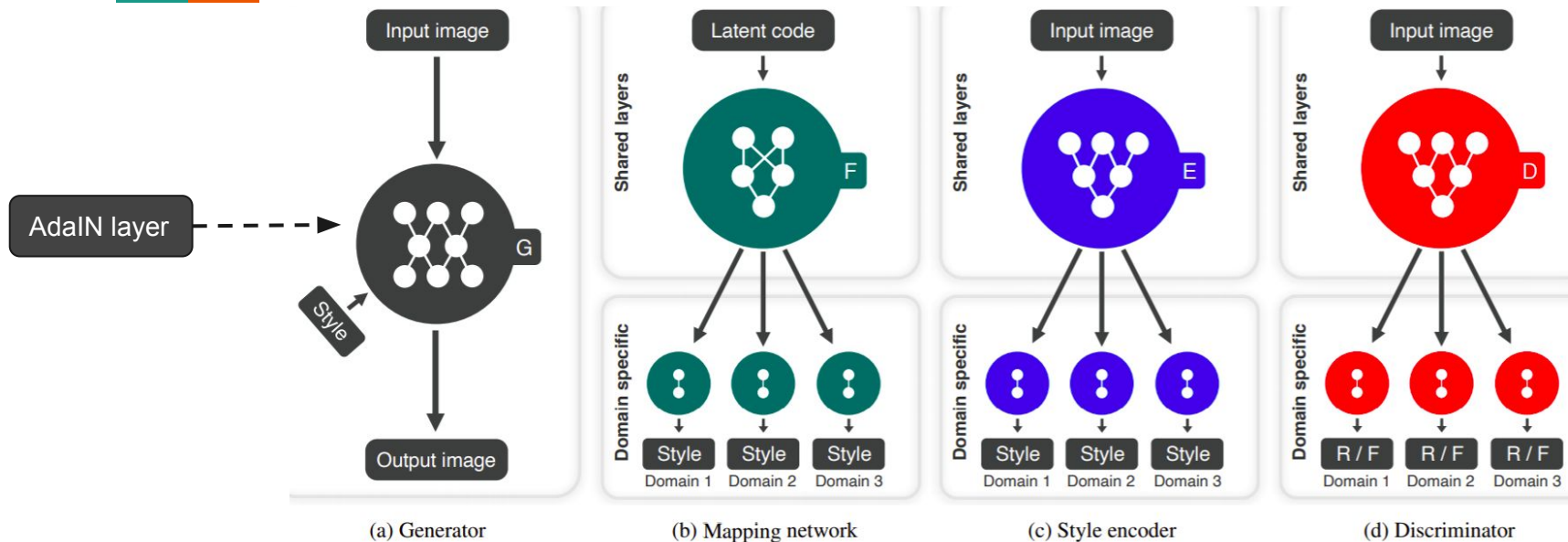


$$\mathcal{L}_{\text{ms}} = \max_G \left(\frac{d_{\mathbf{I}}(G(\mathbf{z}_1, \mathbf{I}), G(\mathbf{I}, \mathbf{z}_2))}{d_{\mathbf{Z}}(\mathbf{z}_1, \mathbf{z}_2)} \right),$$

one-hot domain codes (z_x^d, z_y^d)

$$\begin{aligned} \mathcal{L}_{\text{cls}}^{\text{domain}} = & \mathbb{E}_{x, z_x^d} [-\log D_{\text{cls}}(z_x^d | x)] + \\ & \mathbb{E}_{x, y, z_y^d} [-\log D_{\text{cls}}(z_y^d | G(z_x^c, z_y^a, z_y^d))]. \end{aligned}$$

StarGAN2 (Dec 2019)



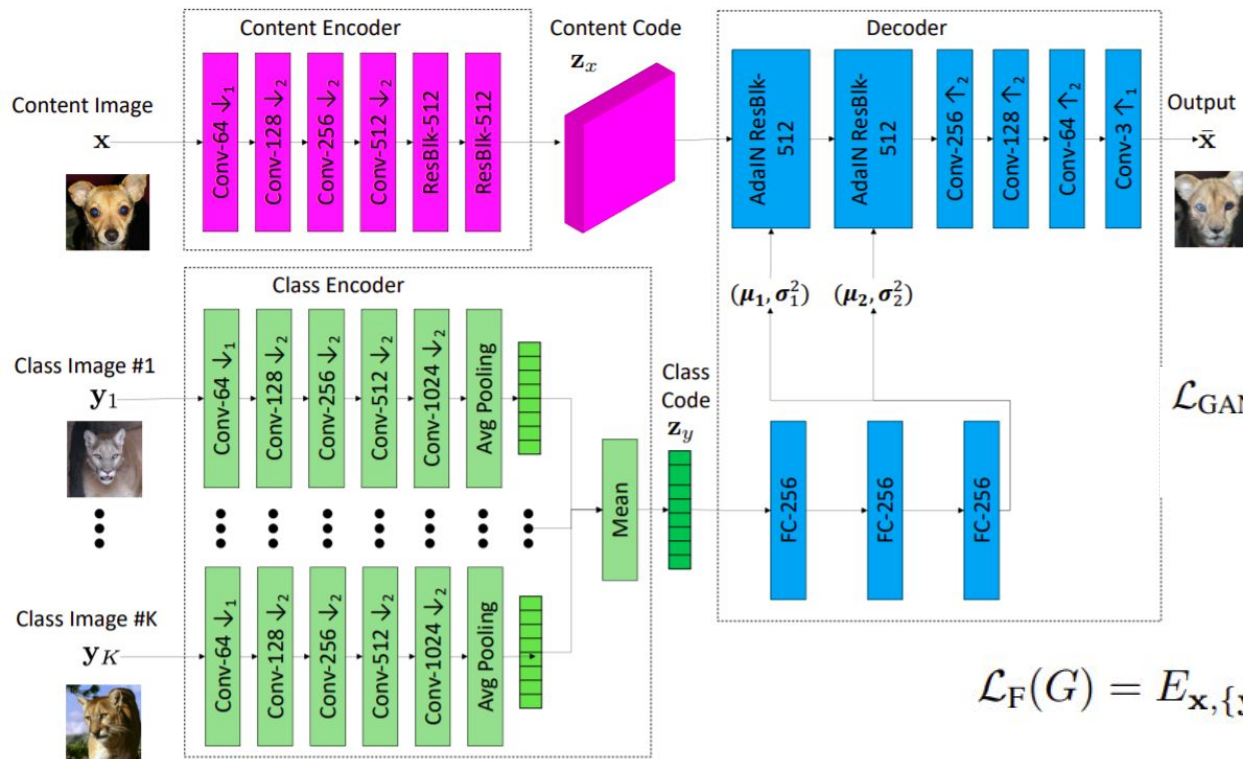
$$\begin{aligned}
 \tilde{\mathbf{s}} &= F_{\tilde{y}}(\mathbf{z}) \\
 \mathcal{L}_{adv} &= \mathbb{E}_{\mathbf{x}, y} [\log D_y(\mathbf{x})] + \\
 &\quad \mathbb{E}_{\mathbf{x}, \tilde{y}, \mathbf{z}} [\log (1 - D_{\tilde{y}}(G(\mathbf{x}, \tilde{\mathbf{s}})))], \quad (1)
 \end{aligned}$$

$$\mathcal{L}_{sty} = \mathbb{E}_{\mathbf{x}, \tilde{y}, \mathbf{z}} [\|\tilde{\mathbf{s}} - E_{\tilde{y}}(G(\mathbf{x}, \tilde{\mathbf{s}}))\|_1]. \quad (2)$$

$$\mathcal{L}_{ds} = \mathbb{E}_{\mathbf{x}, \tilde{y}, \mathbf{z}_1, \mathbf{z}_2} [\|G(\mathbf{x}, \tilde{\mathbf{s}}_1) - G(\mathbf{x}, \tilde{\mathbf{s}}_2)\|_1], \quad (3)$$

$$\mathcal{L}_{cyc} = \mathbb{E}_{\mathbf{x}, y, \tilde{y}, \mathbf{z}} [\|\mathbf{x} - G(G(\mathbf{x}, \tilde{\mathbf{s}}), \hat{\mathbf{s}})\|_1], \quad (4)$$

FUNIT (May 2019)



$$\mathcal{L}_{\text{GAN}}(G, D) = E_{\mathbf{x}} [-\log D^{c_x}(\mathbf{x})] + E_{\mathbf{x}, \{\mathbf{y}_1, \dots, \mathbf{y}_K\}} [\log (1 - D^{c_y}(\bar{\mathbf{x}}))]$$

$$\mathcal{L}_{\text{R}}(G) = E_{\mathbf{x}} [\|\mathbf{x} - G(\mathbf{x}, \{\mathbf{x}\})\|_1^1]$$

$$\mathcal{L}_{\text{F}}(G) = E_{\mathbf{x}, \{\mathbf{y}_1, \dots, \mathbf{y}_K\}} [\|D_f(\bar{\mathbf{x}}) - \sum_k \frac{D_f(\mathbf{y}_k)}{K}\|_1^1]$$

DGNet (Apr 2019)

$$L_{\text{recon}}^{\text{img1}} = \mathbb{E}[\|x_i - G(a_i, s_i)\|_1].$$

$$L_{\text{recon}}^{\text{img2}} = \mathbb{E}[\|x_i - G(a_t, s_i)\|_1].$$

$$L_{\text{recon}}^{\text{code1}} = \mathbb{E}[\|a_i - E_a(G(a_i, s_j))\|_1],$$

$$L_{\text{recon}}^{\text{code2}} = \mathbb{E}[\|s_j - E_s(G(a_i, s_j))\|_1].$$

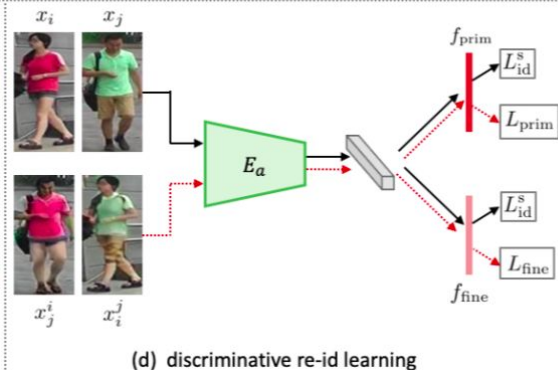
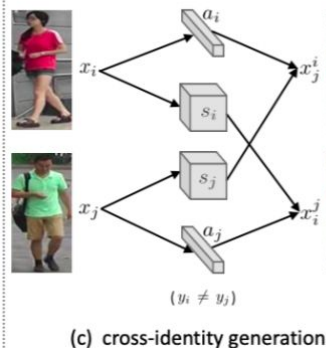
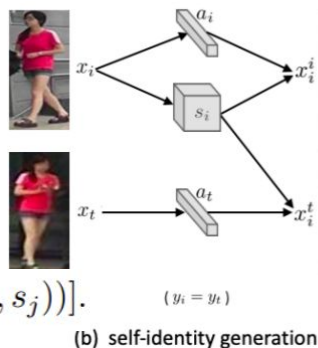
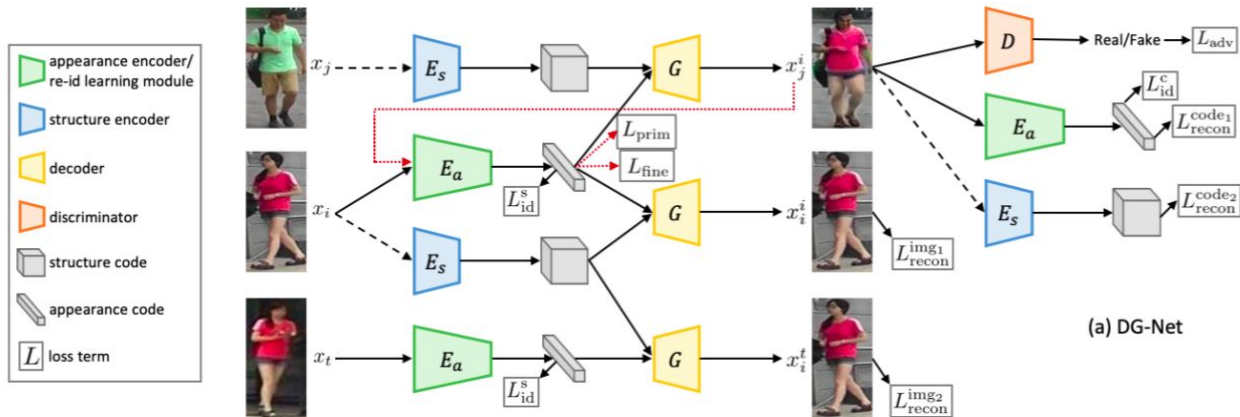
$$L_{\text{prim}} = \mathbb{E}[-\sum_{k=1}^K q(k|x_j^i) \log(\frac{p(k|x_j^i)}{q(k|x_j^i)})],$$

$$L_{\text{fine}} = \mathbb{E}[-\log(p(y_j|x_j^i))].$$

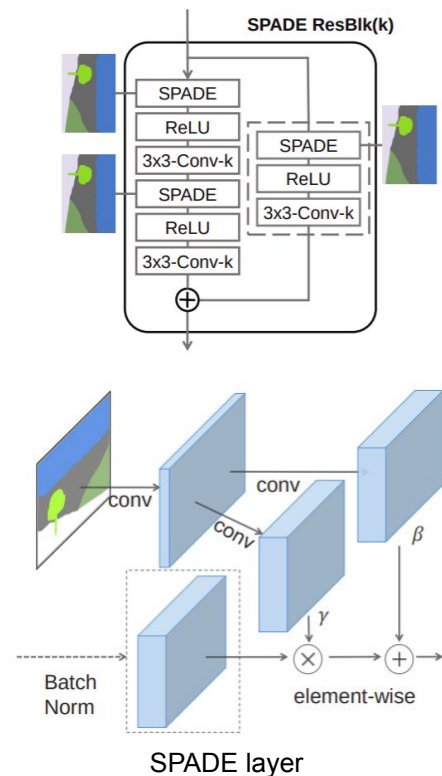
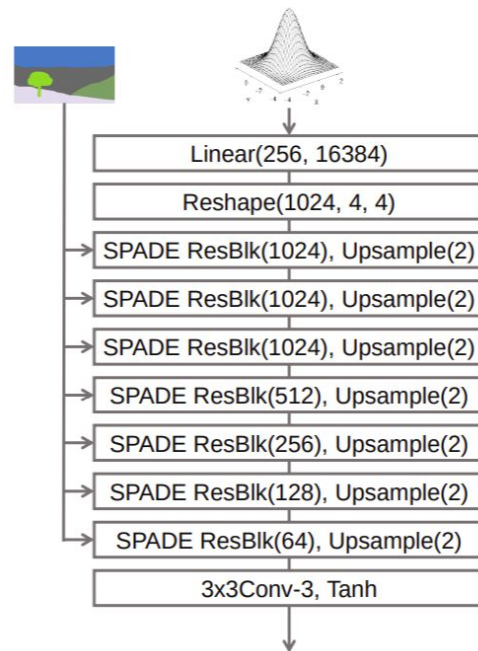
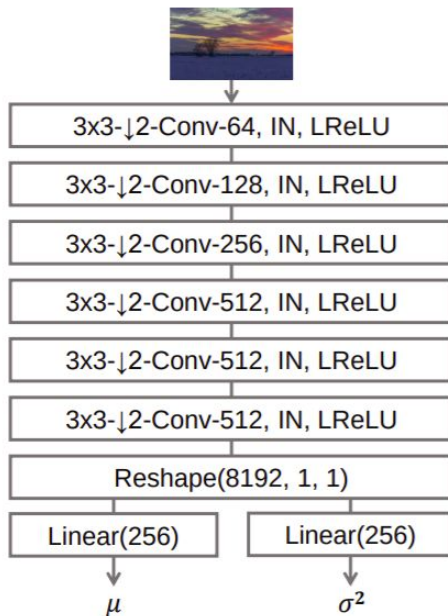
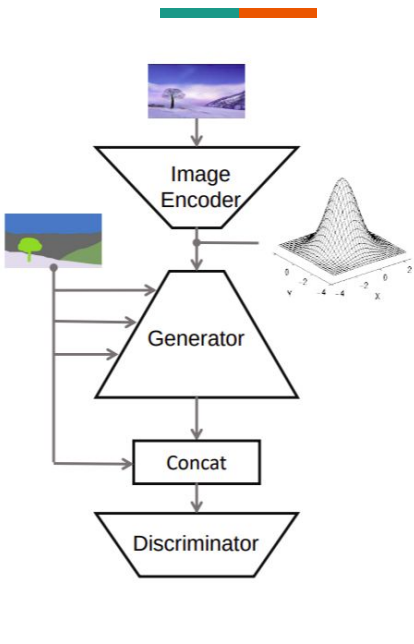
$$L_{\text{id}}^s = \mathbb{E}[-\log(p(y_i|x_i))],$$

$$L_{\text{id}}^c = \mathbb{E}[-\log(p(y_i|x_j^i))],$$

$$L_{\text{adv}} = \mathbb{E}[\log D(x_i) + \log(1 - D(G(a_i, s_j)))].$$



SPADE (Mar 2019)

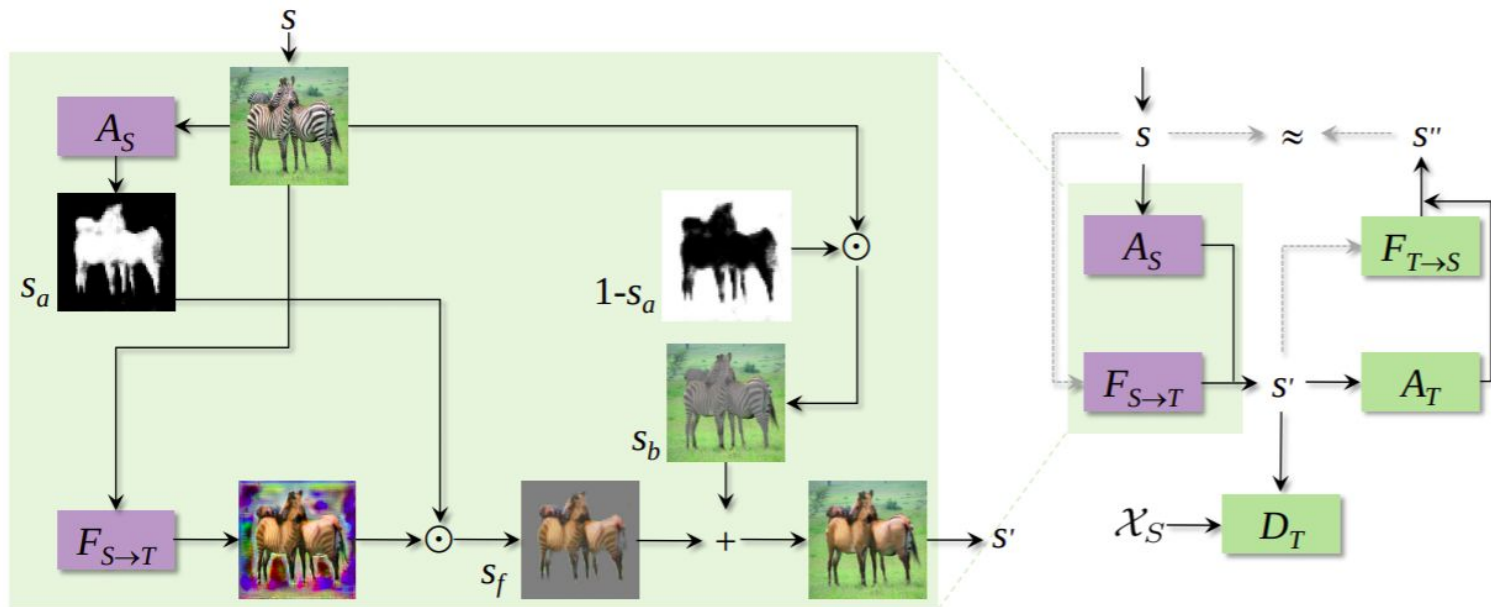


$$\mathcal{L}_{\text{KLD}} = \mathcal{D}_{\text{KL}}(q(\mathbf{z}|\mathbf{x})||p(\mathbf{z}))$$

Author replaced LSGAN by Hinge loss and use KL Divergence loss when training Image Encoder.

Unsupervised Attention-guided Image-to-Image Translation

(Jun 2018)



$$\mathcal{L}_{adv}^s(F_{S \rightarrow T}, A_S, D_T) = \mathbb{E}_{t \sim P_T(t)} [\log(D_T(t))] + \mathbb{E}_{s \sim P_S(s)} [\log(1 - D_T(s'))]$$

$$\mathcal{L}_{cyc}^s(s, s'') = \|s - s''\|_1,$$